Figure 1: BOT screen

Download the BOT binaries from *http://blockouttraffic.de/download.php*. As explained in earlier articles, copy the BOT binaries to the IPCop box using SCP. (Remember that the non-standard IPCop SSH port is 222 TCP.) Log on to the IPCop console locally or via SSH and untar the binaries and install BOT:

```
mkdir BOT
mv ./BlockOutTraffic-3.0.0-GUI-b3.tar.gz ./BOT
cd BOT
tar -xzf BlockOutTraffic-3.0.0-GUI-b3.tar.gz
./setup
```

This completes the basic set-up. Log out from the console/SSH. Further configuration is via the Web-based GUI, so visit it and select *Firewall—Block Outgoing Traffic*. You should see something like what is shown in Figure 1. Click *Edit* to enter the settings menu. Here, you must enter the administrator PC's MAC address and the IPCop proxy port. This is a security feature restricting IPCop Web GUI access only to this PC. Before saving the setting, re-verify both entries; if you enter the wrong values, you will be locked out of the IPCop Web-based GUI. Next, click *Enable BOT*. You should see something like what's shown in Figure 3.

Note: If you are locked out by incorrect MAC/proxy port entries, to regain access, you will require physical console access to the IPCop box, and will then have to run the following commands:

```
iptables -F BOT_INPUT # iptables -F CUSTOMFORWARD
```

After this, access the IPCop Web GUI and correct the MAC address/proxy port. Disable and immediately enable BOT.

Test configuration

Now, let's do a test configuration to allow Internet access as detailed in Table 2. This configuration example assumes that all client browsers are configured to use IPCop proxy port 800.

Figure 2: MAC address and proxy port configuration

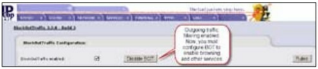

Figure 3: BOT enabled

The first one allows email (POP3 and SMTP) access from the Green network to the outside (the Internet); the other from the Green network to the IPCop proxy (for Web browsing). Go to the administrator GUI's *Firewall—Advanced BOT Config* menu option to create groups and access rules (Figure 4), and then:

1. Create the *MailAccess* list of internal PCs. Under *Address Settings*, configure *MailAccess* as *Name*, *192.168.51.0* as the IP address and 255.255.255.224 (CIDR /27) as the subnet mask.
2. Create the *WebAccess* list of internal PCs. Under *Address Settings*, enter *WebAccess*, *192.168.51.32* and *255.255.255.224* respectively. Figure 5 shows the created rules.
3. Create the *IPCop Proxy* custom service. Under *Services Settings*, enter *Service Name* as *IPCopProxy*, *Port* as 800 (the port defined in *Services—Proxy* or *Advproxy)* and select the *TCP Protocol*.
4. Create the *MailingServices* group. Under *Service Grouping*, supply *MailingServices* as the *Service Group Name*. Select and add the *POP* and *SMTP* services from *Default Services*, as shown in Figure 6.
5. Now, the final step is to create Access Control rules. Proceed to *Firewall—Block Outgoing Traffic* and click on *New Rule*. Create the rule by selecting the following parameters:
   - Source : Default network : Custom Address—MailAccess
   - Destination: Other Network/Outside: Default networks: Any
   - User Services: Service Groups: MailingServices
6. Click *Rule enabled*, add a meaningful remark for this rule,

Table 2: Access matrix

Group name	IP addresses	Services allowed	Service group	Service ports
MailAccess	192.168.51.0/27	POP3, SMTP	*MailingServices*	110 and 25
WebAccess	192.168.51.32/27	Internet access	*InternetServices*	IPCop Proxy